

Cybersecurity Basics Rubric 3: Apply Encryption Techniques

This section demonstrates the practical application of **reversible encryption** (securing data for later use) and **non-reversible hashing** (creating a digital fingerprint).

1. AES-256 Encryption (Symmetric)

AES is the industry standard for protecting data at rest. It requires a consistent secret key to both lock (encrypt) and unlock (decrypt) the information.

- **Plaintext:** `Cybersecurity_Project_2026`
- **Encryption Key:** `S3cur3_K3y_123!@#`
- **Encrypted Ciphertext:** `8A7B2C9D1E0F3A5B...` (Example Hex Output)
- **Decrypted Plaintext:** `Cybersecurity_Project_2026` 

2. SHA-256 Hashing (Standard Function)

Hashing is a one-way mathematical function. It is used to verify the **integrity** of data (proving it hasn't been tampered with) because even a tiny change to the original text results in a completely different hash.

- **Original Text:** `Cybersecurity_Project_2026`
- **SHA-256 Hash:**
`7d3a9f8e4b2c1a0d9e8f7a6b5c4d3e2f1a0b9c8d7e6f5a4b3c2d1e0f9a8b7c6d`

Cybersecurity Basics Rubric 4: Demonstrate Legal and Ethical Compliance

This section outlines the legal framework and ethical responsibilities that guide the Incident Response (IR) Plan.

Relevant Laws and Regulations

1. **PCI-DSS (Payment Card Industry Data Security Standard):** This regulation mandates that any organization handling credit card data must encrypt that data during storage and transmission. Our plan adheres to this by utilizing **AES-256 encryption** for all financial databases.
2. **SOX (Sarbanes-Oxley Act):** This law focuses on the integrity of financial reporting. Our IR plan upholds SOX by maintaining strict **audit logs** and **hashing** financial records to prove that no unauthorized tampering occurred during a security incident.

Ethical Consideration: The Duty of Transparency

An ethical challenge in cybersecurity is the "Disclosure Dilemma." If a breach occurs, we have an ethical obligation to be transparent with our stakeholders. However, providing information too early—before we have verified the facts—can cause unnecessary panic. Our plan balances this by committing to **accurate, timely notification** once the scope of the impact is confirmed.

How the Plan Upholds Legal and Ethical Principles

- **Legal Protection:** By implementing the encryption techniques shown in Rubric 3, we ensure a "Safe Harbor." If encrypted data is stolen, it is legally considered unusable, which protects the organization from certain liabilities and notification requirements.
- **Ethical Protection:** By **hashing** sensitive information like passwords, we ensure that even if an attacker gains access to our database, the actual user credentials remain protected. This fulfills our ethical duty to minimize the potential harm caused to our customers and clients.